```
> restart;#The best fit for photons Planck's low
> E(nu):=Em+E0*ln(1+2.787176283*10^(-17)*nu)+C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.787176283 \times 10^{-17} \nu) + C \nu^2 \quad (1)$$

```
> E1:=subs(nu=9.733521364*10^16,E(nu));
```

$$E1 := Em + E0 \ln(3.712903990) + 9.474143814 \times 10^{33} C \quad (2)$$

```
> E2:=subs(nu=7.880979442*10^14,E(nu));
```

$$E2 := Em + E0 \ln(1.021965679) + 6.210983697 \times 10^{29} C \quad (3)$$

```
> E3:=subs(nu=8.689636465*10^14,E(nu));
```

$$E3 := Em + E0 \ln(1.024219549) + 7.550978189 \times 10^{29} C \quad (4)$$

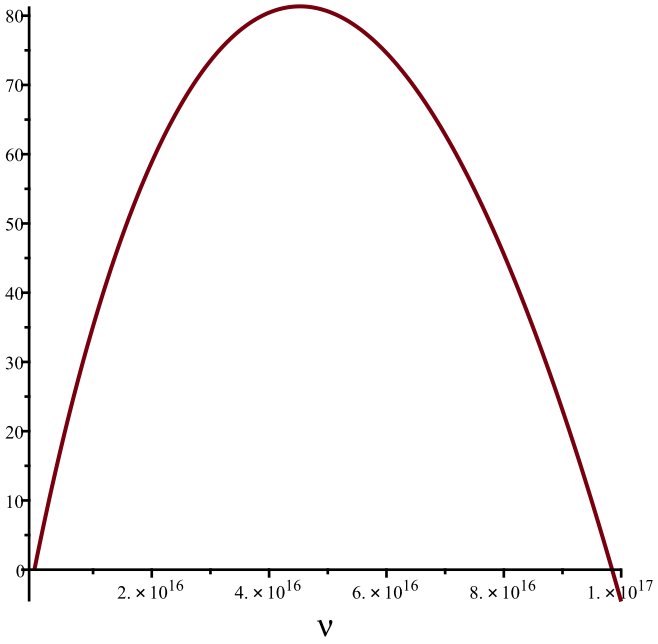
```
> solve({E1=3.259,E2=3.259,E3=3.59});
```

$$\{C = -2.063162888 \times 10^{-32}, E0 = 151.5047392, Em = -0.02006692051\} \quad (5)$$

```
> E(nu):=subs(Em=-0.02006692051,E0=151.5047392,C=-2.063162888*10^(-32),E
  (nu));
```

$$E(\nu) := -0.02006692051 + 151.5047392 \ln(1 + 2.787176283 \times 10^{-17} \nu) - 2.063162888 \times 10^{-32} \nu^2 \quad (6)$$

```
> plot(E(nu),nu=10^13..10^17, caption = typeset("Graph of the Planck's
  constant [eV] as a function of λ in [nm] "));
```



Graph of the Planck's constant [eV] as a function of λ in [nm]

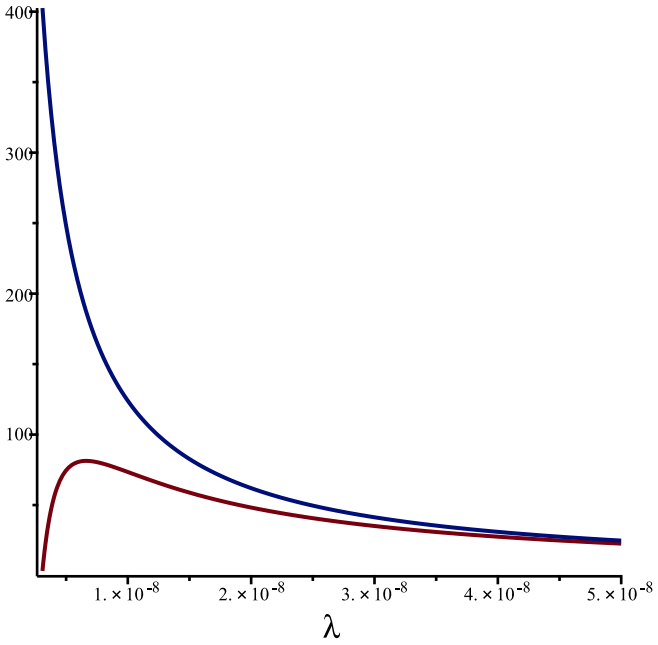
```
> he:=6.6260701500*10^(-34)/1.6021766340/10^(-19);
```

$$he := 4.135667697 \times 10^{-15} \quad (7)$$

```
> c := 299792458;B := lambda -> subs(nu = c/lambda, -0.02006692051 +
  151.5047392*ln(1 + 2.787176283*1/100000000000000000*nu) + (-1)*
  2.063162888*1/10000000000000000000000000000000000*nu^2);plot([B(lambda),
  he*c/lambda], lambda = 3.08*10^(-9) .. 5*10^(-8),caption = typeset("A
  plot of Energy theoretical and practical in [eV] versus lambda in [nm]
  "));
```

$$c := 299792458$$

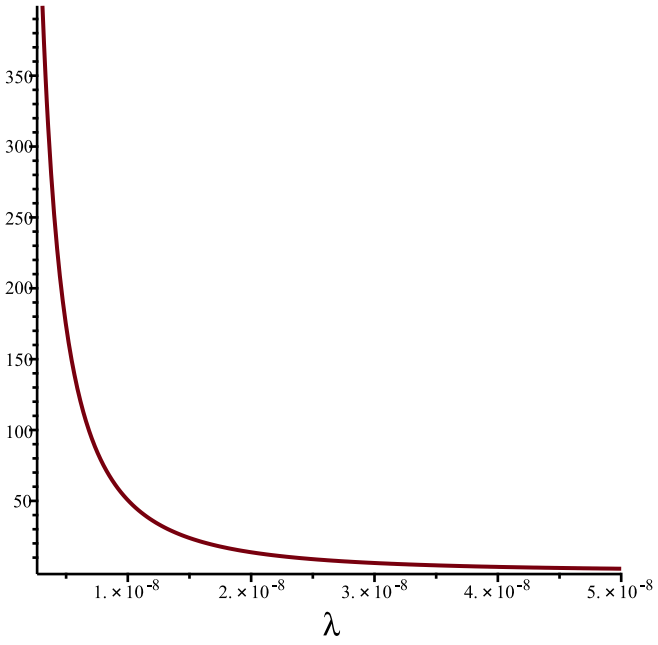
$$B := \lambda \mapsto subs\left(\nu = \frac{c}{\lambda}, -0.02006692051 + 151.5047392 \cdot \ln\left(1 + \frac{2.787176283 \cdot 1 \cdot \nu}{100000000000000000}\right) + \frac{(-1) \cdot 2.063162888 \cdot 1 \cdot \nu^2}{10000000000000000000000000000000000}\right)$$



A plot of Energy theoretical and practical in [eV] versus lambda in [nm]

```
> -B(lambda)+he*c/lambda;plot(%,lambda = 3.08*10^(-9) .. 5*10^(-8),
  caption = typeset("Graph of the error [eV] as a function of λ in [nm]
  between the two calculation methods "));# Below the wavelength of 30
  nm, the error becomes larger and below 15-20nm the error is dominant.
```

$$0.02006692051 - 151.5047392 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) + \frac{1.854278330 \times 10^{-15}}{\lambda^2} + \frac{1.239841984 \times 10^{-6}}{\lambda}$$



Graph of the error [eV] as a function of λ in [nm] between the two calculation methods

```
> solve(diff(B(lambda), lambda) = 0);# @ lambda= 6.625nm the photons
  have maximum of energy
```

$$-3.695060778 \times 10^{-9}, 6.624561165 \times 10^{-9} \quad \textbf{(8)}$$

```
> h := 6.6260701500*10^(-34);ec := 1.6021766340*10^(-19);rb:=5.239*10^
  (-13);he:=h/ec;
  c := 299792458;T := 273 + 24;k := 1.3806490000*10^(-23);Tq:= 1.765*10^
  (-19);nu0:=evalf(c/(3.804*10^(-7)));nu1:=evalf(c/(3.08*10^(-9)));nu2:=
  evalf(c/(3.135*10^(-7)))
```

$$h := 6.626070150 \times 10^{-34}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$rb := 5.239000000 \times 10^{-13}$$

$$he := 4.135667697 \times 10^{-15}$$

$$c := 299792458$$

$$T := 297$$

$$k := 1.380649000 \times 10^{-23}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$v0 := 7.880979442 \times 10^{14}$$

$$v1 := 9.733521364 \times 10^{16}$$

$$v2 := 9.562757830 \times 10^{14} \quad (9)$$

```
> evalf(subs(nu=7.880979442*10^14,E(nu)));
```

$$3.258999999 \quad (10)$$

```
> evalf(subs(nu=8.689636465*10^14,E(nu)));
```

$$3.590000000 \quad (11)$$

```
> evalf(subs(nu=9.733521364*10^16,E(nu)));
```

$$3.2590000 \quad (12)$$

```
> evalf(subs(nu=nu2,E(nu)));
```

$$3.946259923 \quad (13)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=5*10^12,E(nu)));evalf(subs(nu=
  5.2*10^16,E(nu)));evalf(c/(5*10^12));#The  max energy ~80eV wavelenght
  60 microns÷3 nm;
```

$$-0.02006692051$$

$$0.00104464160$$

$$79.91225659$$

$$0.00005995849160 \quad (14)$$

```
> solve(-0.02006692051 + 151.5047392*ln(1 + 2.787176283*10^(-17)*nu) -
  2.063162888*10^(-32)*nu^2=5.28,nu)
Warning, solutions may have been lost
```

```
> evalf(subs(nu=5*10^16,E(nu)));
```

$$80.63322078 \quad (15)$$

```
> evalf(subs(nu=0.984563*10^17,E(nu)));c/(0.984563*10^17);
```

$$0.0000560$$

$$3.044929150 \times 10^{-9} \quad (16)$$

```
> 1238.4/8.54;nu3:=c/(%*10^(-9));
```

$$145.0117096$$

$$v3 := 2.067367241 \times 10^{15} \quad (17)$$

```
> evalf(subs(nu=nu3,E(nu)));8.54-%;# @ 145nm the error is 0.16eV;
```

$$8.379383055$$

$$0.160616945 \quad (18)$$

```
> 1238.4/5.54;nu3:=c/(%*10^(-9));
```

$$223.5379061$$

$$v3 := 1.341125822 \times 10^{15} \quad (19)$$

```
> evalf(subs(nu=nu3,E(nu)));5.54-%;#@ 223nm the error is 0.037eV;
```

$$5.502724559$$

$$0.037275441 \quad (20)$$

```
> plot([E(nu),he*nu], nu=10^13..5*10^16);plot([E(nu),he*nu], nu=10^13.
  .10^17);
```

